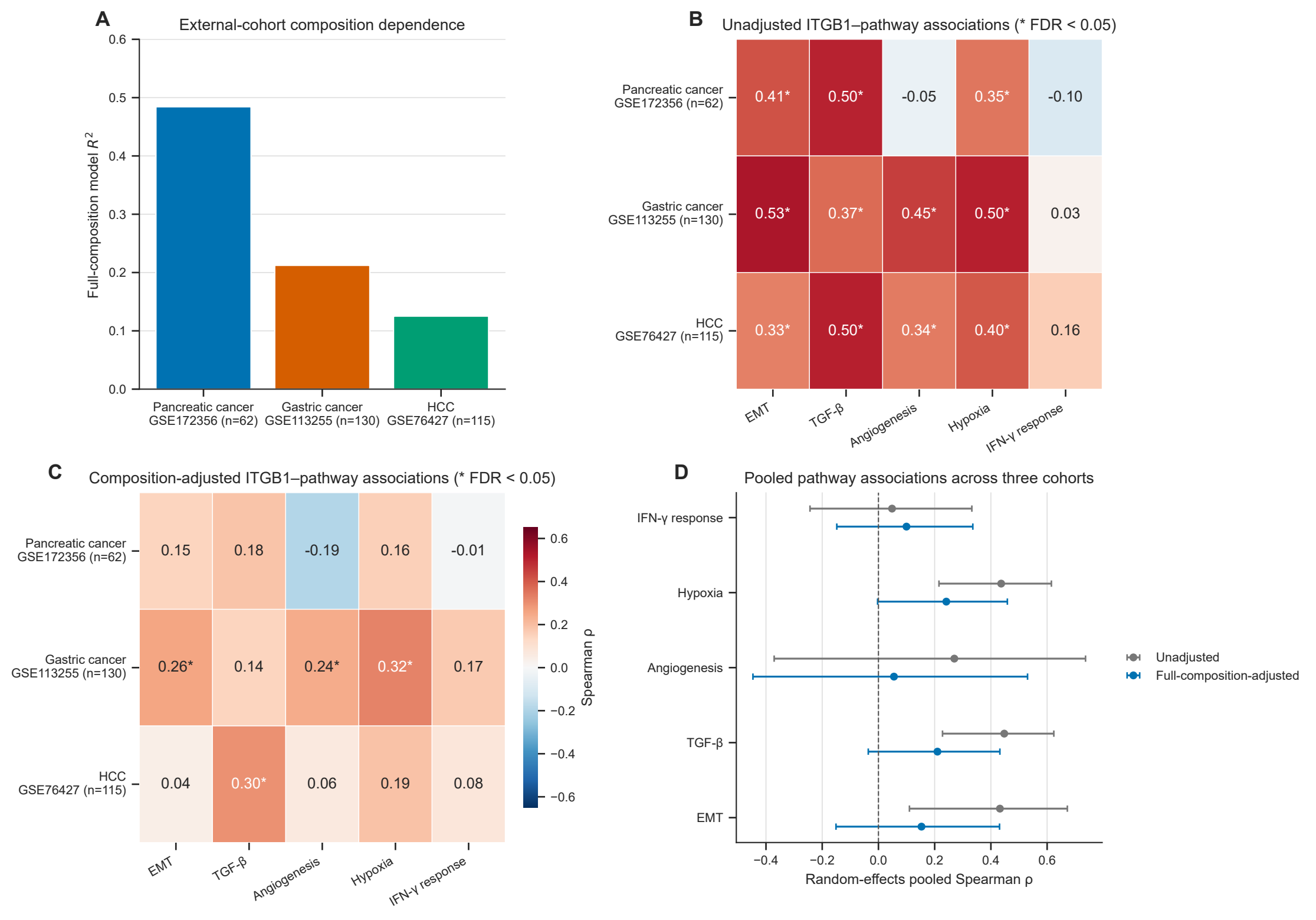

A
External-cohort composition dependence
Full-composition model R^2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Pancreatic cancer GSE172356 (n=62)
Gastric cancer GSE113255 (n=130)
HCC GSE76427 (n=115)
B
Unadjusted ITGB1–pathway associations (* FDR < 0.05)
Pancreatic cancer GSE172356 (n=62)
Gastric cancer GSE113255 (n=130)
HCC GSE76427 (n=115)
0.41* 0.50* -0.05 0.35* -0.10
0.53* 0.37* 0.45* 0.50* 0.03
0.33* 0.50* 0.34* 0.40* 0.16
EMT
TGF-β
Angiogenesis
Hypoxia
IFN-γ response
C
Composition-adjusted ITGB1–pathway associations (* FDR < 0.05)
Pancreatic cancer GSE172356 (n=62)
Gastric cancer GSE113255 (n=130)
HCC GSE76427 (n=115)
0.15 0.18 -0.19 0.16 -0.01
0.26* 0.14 0.24* 0.32* 0.17
0.04 0.30* 0.06 0.19 0.08
EMT
TGF-β
Angiogenesis
Hypoxia
IFN-γ response
Spearman ρ
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
D
Pooled pathway associations across three cohorts
IFN-γ response
Hypoxia
Angiogenesis
TGF-β
EMT
−0.4
−0.2
0.0
0.2
0.4
0.6
Random-effects pooled Spearman ρ
Unadjusted
Full-composition-adjusted